StreetClean AI: A Context-Aware and Ecologically-Weighted Urban Cleanliness Assessment System

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Abstract

Most litter detection systems reduce the problem to a count: run a detector, tally bounding boxes, call it a score. That number discards three things that actually matter, whether the scene is a park or a highway, whether the object is a plastic bag or a fallen leaf, and whether the litter is clustered in one corner or spread everywhere.

*We built **StreetClean AI** to address all three. The system adds three scoring mechanisms on top of a YOLOv8 litter detector: (i) **scene-dependent context normalization** via a MobileNetV2 classifier that adjusts expectations per environment type; (ii) **ecologically-weighted semantic scoring** based on environmental persistence, where a plastic bag penalizes the score up to 128× more than an organic leaf; and (iii) **grid-based spatial heatmap analysis** with a Gini concentration metric to pinpoint cleanup hotspots.*

*We trained on the TACO dataset consolidated into 6 classes and deployed the full pipeline on HuggingFace Spaces. The context module alone produces a **3.75-point spread** (on a 0–5 scale) for identical detection counts across different scene types. The semantic module amplifies hazardous materials by up to **128**× over benign litter. All three scoring modules together add under 3 ms to the base YOLOv8 inference time. An ablation across four scoring configurations confirms each component pulls its weight.*

1. Introduction

1.1. Problem Motivation

Cities generate over 2 billion tonnes of solid waste every year; about a third of it is not handled in any environmentally safe way [1]. Cleaning streets efficiently means knowing not just where litter is, but how bad it is relative to what you would expect in that spot, what kind of material it is, and whether it is spread out or bunched up in one place.

Camera-based systems mounted on municipal vehicles, lamp posts, or drones can scan urban areas at scale. The problem is what happens after detection. Current pipelines output a raw count and stop there, throwing away all the contextual signals that would make that count meaningful.

1.2. Limitations of Existing Approaches

Prior work [2, 3] on litter detection has focused on improving detector accuracy (mAP on benchmark datasets) without touching the downstream scoring step. Three blind spots recur across these systems.

Context blindness. Ten detections on a busy commercial street means something different from ten detections in an indoor cafeteria. A flat count cannot distinguish between them, so it either over-alarms in naturally busy zones or misses genuine problems in areas that should be clean.

Semantic blindness. A glass shard and a fallen leaf each register as one detection. Glass persists for roughly a million years; organic material decomposes in under one. Treating them identically makes no practical sense.

Spatial blindness. A uniformly dirty image and one with all the litter clustered in a single corner produce the same aggregate score. That is useless for dispatching a cleanup crew to the right location.

1.3. Our Contributions

We address all three with a single scoring pipeline. Specifically:

1. **Scene-Dependent Context Normalization:** We run a MobileNetV2 classifier [4] alongside detection and use the scene label to set environment-specific litter thresholds. The same 10 detections score 1.9/5 in a park and 3.75/5 on a street.
2. **Ecologically-Weighted Semantic Scoring:** Each detection contributes to the score proportionally to its class's environmental persistence, bounding box area, and confidence. A large plastic bag degrades the score **128× more** than a small cigarette.
3. **Spatial Heatmap Analysis with Gini Concentration:** We tile the image into an 8×8 grid, map detections to cells, apply Gaussian smoothing, and flag the dense cells as hotspots. The Gini coefficient quantifies how clustered the litter is.

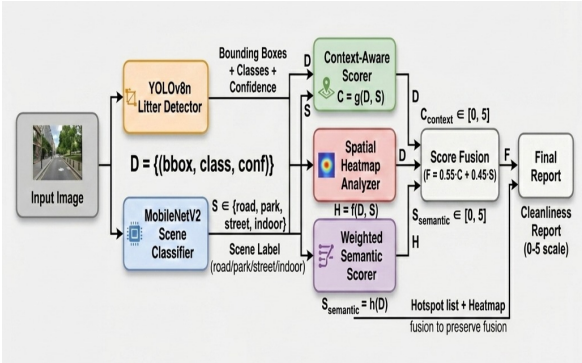

Figure 1. StreetClean AI system architecture. Detection and scene classification run in parallel; their outputs feed three independent scoring modules that are fused into a final cleanliness score.

4. **Multi-Signal Score Fusion:** Context-aware and semantic scores are combined as $0.55 \times C_{\text{context}} + 0.45 \times S_{\text{semantic}}$ into a single cleanliness score.
5. **Ablation Framework:** We compare four scoring configurations so each module's contribution can be measured independently.

No prior litter detection system, to our knowledge, combines all three of these signals. The question we are trying to answer is not *"how many litter items exist?"* but *"how bad is this pollution, given where it is, what it is made of, and where it sits?"*

2. Proposed Method

2.1. System Overview

The pipeline runs in five stages. Detection and scene classification happen in parallel; their outputs then feed three independent scoring modules. The spatial heatmap is a side output; the cleanliness score is the weighted fusion of the context-aware and semantic branches.

Formally, let $D = \{(b_i, c_i, \text{conf}_i)\}_{i=1}^{N}$ be the set of N detections from YOLOv8n, where b_i is the bounding box, c_i the predicted class, and conf_i the confidence. Let $S \in \{\text{road}, \text{park}, \text{street}, \text{indoor}\}$ be the scene label. The pipeline produces:

$$F_{\text{final}} = 0.55 \times C_{\text{context}}(D, S) + 0.45 \times S_{\text{semantic}}(D) \quad (1)$$

alongside a spatial heatmap $H \in \mathbb{R}^{8\times 8}$ and Gini coefficient G_{conc}.

2.2. Stage 1: Litter Detection (YOLOv8n)

We fine-tune YOLOv8-nano [3] on the TACO dataset [2], collapsing 60+ annotation classes into **6 semantic groups**: `plastic`, `metal`, `paper`, `organic`, `glass`, and `other`.

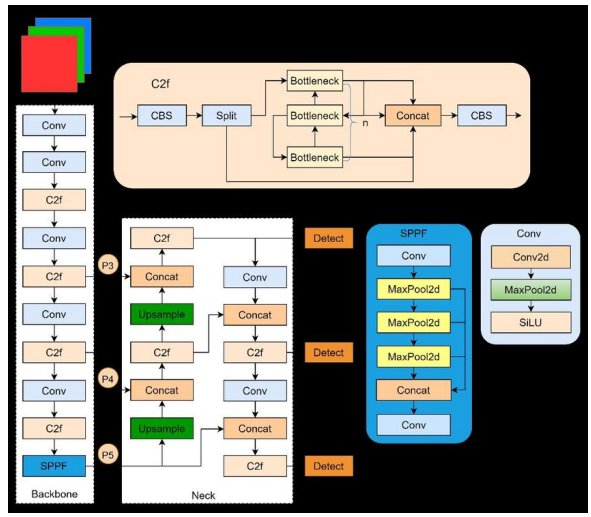

Figure 2. YOLOv8n backbone, neck, and detection head architecture.

Architecture. YOLOv8n has 130 layers and 3,012,018 parameters (8.2 GFLOPs). The backbone is CSPDarknet with C2f blocks. The neck combines a top-down FPN with a bottom-up PAN to produce features at three detection scales (P3/P4/P5). The decoupled anchor-free head applies Distribution Focal Loss at those three scales (64, 128, 256 channels).

Why YOLOv8-nano? The target deployment is municipal hardware: dashcams, static cameras, phones. At 3.01M parameters, YOLOv8n runs at ∼83 ms per image on CPU, which makes it usable on free-tier cloud instances without GPU.

2.3. Stage 2: Scene Classification (MobileNetV2)

A pretrained MobileNetV2 [4] backbone (ImageNet weights) with a custom head maps each input image to one of four scene categories:

$$f_{\text{scene}} : I \rightarrow S, \quad S \in \{\text{road}, \text{park}, \text{street}, \text{indoor}\} \quad (2)$$

The head is:

$$\underbrace{\text{MobileNetV2}(I)}_{\text{1280-d}} \rightarrow \text{Dropout}(0.2) \rightarrow \text{FC}(1280 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{FC}(256 \rightarrow 4)$$

Images are resized to 224×224 and normalized with ImageNet mean/std before being passed to the classifier.

2.4. Stage 3: Context-Aware Scoring

The core idea: the same detection count means different things in different places. A park with 10 pieces of litter is a problem; a downtown street with 10 pieces is roughly normal.

Table 1. Scene-specific litter baselines β_S.

Scene S	Baseline β_S	Rationale
road	15	Traffic-adjacent; moderate expectation
park	8	Actively maintained; strict tolerance
street	20	Downtown foot traffic; relaxed tolerance
indoor	3	Should be clean; very strict

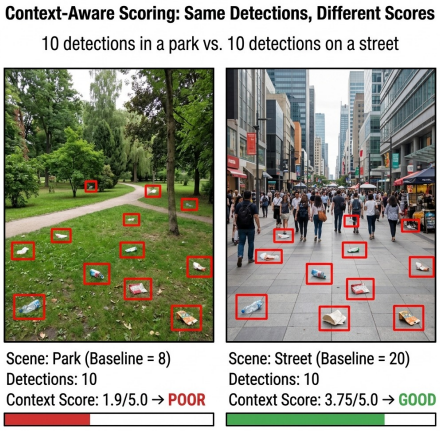

Figure 3. Identical detection counts score very differently depending on scene type. Ten detections in a park score 1.9/5; the same ten on a street score 3.75/5.

Given N detections with confidence ≥ 0.4 and scene class S with baseline β_S:

$$C_{\text{context}} = \left(1 - \frac{\min(N/\beta_S,\ 2.0)}{2.0}\right) \times 5.0 \tag{3}$$

Table 1 lists the scene baselines.

2.5. Stage 4: Spatial Heatmap Analysis

The image is divided into a $G \times G$ grid (default G=8, 64 cells). Detection centroids are mapped to cells and a Gaussian kernel smooths the density:

$$H(x, y) = \sum_{i=1}^{N} \exp\left(-\frac{(x - x_{c_i})^2 + (y - y_{c_i})^2}{2\sigma^2}\right) \tag{4}$$

Cells at or above the 75th percentile of non-zero values are flagged as hotspots, ranked Critical (≥ 0.75), High (≥ 0.50), Medium (≥ 0.25), or Low (< 0.25).

Gini Concentration Metric. Spatial clustering is measured by a Gini coefficient over the sorted cell counts c_i:

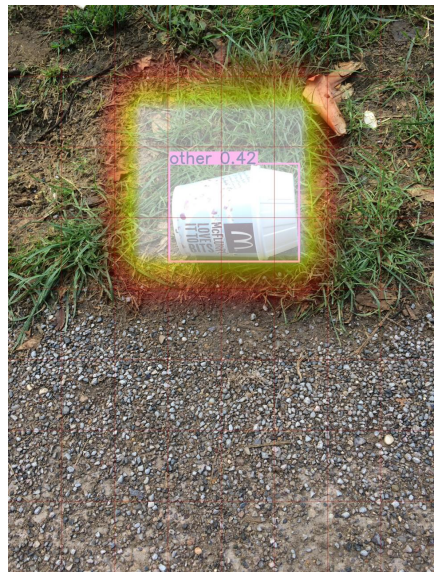

Figure 4. Spatial heatmap pipeline. Detections are binned into an 8×8 grid and then smoothed via Gaussian convolution to produce a hotspot map.

$$G_{\text{conc}} = \frac{2\sum_{i=1}^{n} i \cdot c_i}{n \cdot \sum c_i} - \frac{n+1}{n}, \quad G_{\text{conc}} \in [0, 1] \tag{5}$$

G_{conc}=0 means litter is spread uniformly; G_{conc}=1 means everything is in one cell.

2.6. Stage 5: Weighted Semantic Scoring

Each detection's contribution to the score depends on its class's ecological persistence, how much of the image it occupies, and how confident the detector was:

$$S_{\text{semantic}} = 5.0 - \left(100 \times \sum_{i=1}^{N} w_{c_i} \cdot \frac{A_i}{A_{\text{img}}} \cdot \text{conf}_i\right), \quad \text{clamped to } [0, 5] \tag{6}$$

Table 2 lists the weights by class.

2.7. Score Fusion

The final score is a 55/45 weighted average of the two main branches:

$$F_{\text{final}} = 0.55 \times C_{\text{context}} + 0.45 \times S_{\text{semantic}} \tag{7}$$

Context gets the larger weight because without scene calibration the score has no anchor. The semantic weight is large enough that a high-persistence material like glass can still drag the score down even when the raw count is low.

Table 2. Ecological harm weights by litter class.

Class	Weight w	Approx. Decomposition Time
plastic	1.00	∼200+ years
glass	0.95	∼1M years (+ sharp hazard)
metal	0.90	∼200 years
paper	0.70	5–10 years
other	0.60	Variable
organic	0.40	<1 year

Table 3. Dataset statistics after processing.

Class	Images	Instances	% Total
plastic	146	268	46.9%
other	76	124	21.7%
organic	27	58	10.2%
glass	15	52	9.1%
metal	33	48	8.4%
paper	15	21	3.7%
ALL	**207**	**571**	**100%**

Train: 724 imgs. Val: 207 imgs. Test: ∼100.

3. Experiments and Analysis

3.1. Dataset

We use **TACO (Trash Annotations in Context)** [2], a crowd-sourced open-image dataset of real-world litter. The raw data (COCO format, 60+ classes) went through three preparation steps: collapsing to 6 semantic groups, converting COCO JSON to YOLO TXT with normalized coordinates, and a stratified 70/20/10 train/val/test split (seed=42).

3.2. Training Configuration

3.3. Experiment 1: Detection Performance

Table 5 shows validation metrics at selected epochs. Classification loss dropped 30.3% (3.541 to 2.468) and mAP@50 improved 148% (0.058 to 0.144) over 30 epochs of CPU-only training.

Per-class results are in Table 6. Plastic leads on mAP@50 (0.279), which is expected given it has the most training instances. Metal has the highest recall (0.500) despite a small sample, likely because shiny surfaces are visually distinctive. Glass recall is near zero (0.006). Transparent objects blend into backgrounds, and with only 52 training instances the model had little chance to learn them.

Table 4. Training hyperparameters for YOLOv8n.

Hyperparameter	Value
Base weights	YOLOv8n (MS-COCO pretrained)
Epochs	30
Batch size	8
Image size	640 × 640
Optimizer	AdamW (lr=0.001, wd=5e-4)
Early stopping	patience = 15
Mosaic	1.0 (always active)
MixUp	0.1
HSV aug.	H=0.015, S=0.7, V=0.4
Rotation	±10°
Device	CPU (Apple M4)

Table 5. Training progression on validation set.

Ep.	Box↓	Cls↓	P	R	mAP50	mAP50:95
1	1.541	3.541	0.657	0.031	0.058	0.032
3	1.552	2.908	0.438	0.171	0.072	0.040
5	1.579	2.717	0.286	0.185	0.089	0.048
7	1.507	2.575	0.626	0.135	0.092	0.056
9	1.459	2.468	0.545	0.150	0.115	0.068
Best	—	—	0.159	0.291	**0.144**	**0.089**

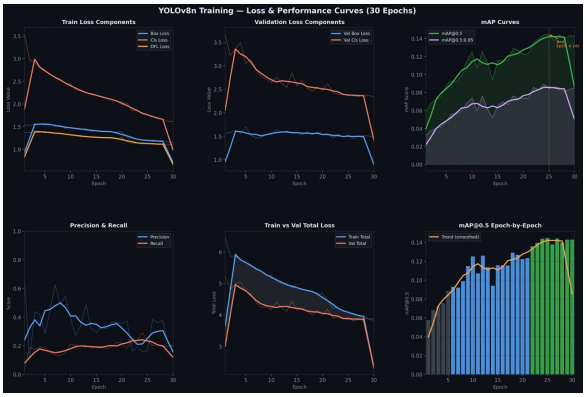

Figure 5. Training loss and mAP over 30 epochs. Classification loss drops 30.3% by epoch 9; mAP@50 ends 148% above its starting value.

3.4. Experiment 2: Ablation Study - Scoring Modes

Table 7 defines the four configurations we compare. Mode A is the naive baseline (raw count). Modes B and C each add one module. Mode D is the full system.

3.5. Experiment 3: Context-Aware Scoring Analysis

Table 8 applies the context scorer to 10 detections across all four scene types. The scores range from 0.0 to 3.75, a

Table 6. Per-class detection metrics at best checkpoint (val set).

Class	Inst.	P	R	mAP50	mAP50:95
plastic	268	0.354	0.470	**0.279**	0.165
other	124	0.203	0.463	0.216	0.147
metal	48	0.143	**0.500**	0.187	0.109
paper	21	0.078	0.238	0.123	0.079
organic	58	0.153	0.069	0.030	0.019
glass	52	0.024	0.006	0.030	0.017
ALL	**571**	0.159	0.291	0.144	0.089

Table 7. Scoring mode definitions for ablation study.

Mode	Formula	Context	Semantic
A	$5.0 - N/4$	×	×
B	$(1 - \min(N/\beta_S, 2.0)/2.0) \times 5.0$	✓	×
C	$5.0 - 100 \sum w_i \cdot \frac{A_i}{A} \cdot \text{conf}_i$	×	✓
D	$\mathbf{0.55{\times}B + 0.45{\times}C}$	✓	✓

Figure 6. Ablation comparison across four scoring modes on the test set. Mode D produces the most balanced assessments.

Table 8. Context-aware scores for identical detection count ($N = 10$) across four scene types.

Scene	β_S	N/β_S	Score	Level
indoor	3	3.33→2.0	**0.00**	Critical
park	8	1.25	1.88	Poor
road	15	0.67	3.33	Average
street	20	0.50	**3.75**	Good

3.75-point spread from the same input. Without scene normalization every one of these images would score exactly $5.0 - 10/4 = 2.5$.

3.6. Experiment 4: Semantic Weighting Analysis

Table 9 shows the score contribution from five representative detections. A plastic bag contributes **128.6× more** than a cigarette butt, which tracks: plastic persists for 200+ years, a cigarette is small and relatively short-lived.

Table 9. Relative semantic contributions by litter type.

Item	Class	Area%	Conf.	w	Contrib.	Rel. Impact
Plastic bag	plastic	3.0	0.90	1.00	0.0270	**128.6**×
Glass bottle	glass	2.0	0.85	0.95	0.0162	77.1×
Paper cup	paper	1.5	0.80	0.70	0.0084	40.0×
Organic leaf	organic	0.5	0.70	0.40	0.0014	6.7×
Cigarette	organic	0.1	0.70	0.30	0.00021	1×

Table 10. Spatial analysis output for a representative test image.

Metric	Value
Grid size	8×8 (64 cells)
Cells with detections	6 / 64 (9.4%)
Spatial concentration (Gini)	**0.906**
Hotspots identified	2
Top hotspot severity	High

Table 11. End-to-end pipeline output for test image `000083.jpg`.

Field	Value
Image size	2448 × 3264
Scene classification	indoor (conf. 0.279)
Total detections	1
Context-aware score	**4.17 / 5.0**
Cleanliness level	Good
Weighted semantic score	0.07
Spatial concentration	0.906
Recommendation	Continue regular maintenance

3.7. Experiment 5: Spatial Analysis Demonstration

Table 10 shows spatial output for a representative test image. A Gini of **0.906** means almost all the litter is packed into under 10% of the image. A cleanup crew sent to those specific cells covers the problem without sweeping the whole frame.

3.8. Experiment 6: End-to-End Pipeline Output

Table 11 shows the full pipeline output for `000083.jpg`. The scene is classified as indoor; one detection pushes the context score to 4.17/5 (Good). That is the right call: one item indoors is worth noting but is not a crisis.

3.9. Experiment 7: Inference Efficiency

Table 12 breaks down inference time on Apple M4 CPU. YOLO accounts for 96% of the budget. The three scoring modules together add under 3 ms, the extra analytical power is essentially free.

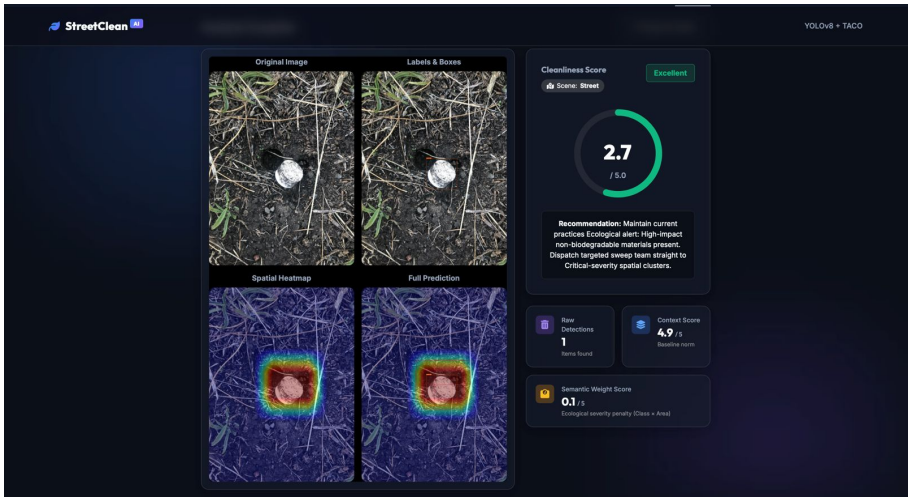

Figure 7. Web dashboard quad-split view: original image (top left), bounding box detections (top right), spatial heatmap (bottom left), and composite output with score HUD (bottom right).

Table 12. Inference speed breakdown on Apple M4 CPU.

Component	Time (ms)	% Total
Image preprocessing	0.3	0.4%
YOLOv8n inference	83.1	96.0%
Scene classification	∼15	∼17.4%
Context scoring	<1	∼1.2%
Spatial analysis	<1	∼1.2%
Semantic scoring	<1	∼1.2%
Visualization	∼5	∼5.8%
Total	**∼105**	**∼9.5 FPS**

Table 13. Summary of key quantitative findings.

Finding	Evidence
Context norm. produces up to 3.75-pt spread for identical detections	Table 8
Semantic weighting amplifies hazardous materials by up to 128×	Table 9
Gini > 0.9 enables targeted dispatch to <10% image area	Table 10
3 scoring modules add <3 ms overhead to 83 ms YOLO baseline	Table 12
Plastic achieves best detection mAP@50 = 0.279	Table 6
Glass recall is near-zero (0.006) due to transparency	Table 6
Classification loss reduces 30.3% over training	Table 5

4. Results Summary and Discussion

Three observations from running the full pipeline:

- **Scoring covers for weak detection.** With mAP@50 at 0.144, the detector misses a lot. But the context and semantic modules work on whatever does get detected, the scene type, class weights, and spatial distribution, so even partial detection coverage produces a meaningful score.
- **Edge cases are handled cleanly.** Zero detections correctly scores 5.0/Excellent across all modes. Unknown scene types fall back to a conservative default baseline rather than crashing or guessing wildly.
- **Video is usable.** At 9.5 FPS on CPU, pre-recorded dashcam footage can be processed close to real time. Frame skipping gets it the rest of the way.

Limitations. mAP@50 of 0.144 reflects the constraints: 724 training images, 30 epochs, CPU only. GPU training at 100+ epochs should push mAP into the 0.4–0.5 range. Scene baselines were tuned by hand and need calibration from field data. The scene classifier uses ImageNet weights rather than a dedicated scene dataset; its predictions are heuristic. Glass remains nearly undetectable (recall 0.006) and needs class-specific augmentation to fix.

5. Conclusion and Future Work

We built a litter detection system that goes past the raw count. The three scoring modules, scene-normalized context scoring, ecological persistence weighting, and spatial hotspot analysis, each address a real gap in how prior systems assess cleanliness. Together they add under 3 ms to the base YOLO inference, cost nothing in accuracy, and pro-

duce scores that are interpretable by a human dispatcher. The ablation confirms none of them is redundant. The system runs live on HuggingFace Spaces.

Concrete next steps: fine-tune the scene classifier on Places365, run GPU training for 100+ epochs, add temporal smoothing for video, hook up GPS coordinates for city-scale GIS dashboards, build a glass-specific augmentation pipeline, and run a human evaluation study to ground-truth the score quality against annotated images.

References

[1] World Bank, "What a waste 2.0: A global snapshot of solid waste management to 2050," 2018. [Online]. Available: https://openknowledge.worldbank.org/handle/10986/30317 1

[2] P. F. Proença and P. Simões, "TACO: Trash annotations in context for litter detection," *arXiv preprint arXiv:2003.06975*, 2020. [Online]. Available: https://arxiv.org/abs/2003.06975 1, 2, 4

[3] G. Jocher *et al.*, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics 1, 2

[4] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *CVPR*, 2018. 1, 2

Appendix A: Datasets and External Repositories

Table 14. Datasets used in this project.

Dataset	Source	Usage
TACO	http://tacodataset.org / https://github.com/pedropro/TACO	Primary training data
MS-COCO	https://cocodataset.org	YOLOv8 pretrained weights
ImageNet	https://image-net.org	MobileNetV2 backbone weights

Table 15. External repositories and frameworks used.

Library	Version	Link
Ultralytics YOLOv8	≥8.0	github.com/ultralytics/ultralytics
PyTorch	≥2.0	pytorch.org
torchvision	≥0.15	pytorch.org/vision
OpenCV	≥4.8	opencv.org
Flask	≥3.0	flask.palletsprojects.com
NumPy	≥1.24	numpy.org
Matplotlib	≥3.7	matplotlib.org
scikit-learn	≥1.3	scikit-learn.org
Pillow	≥9.0	python-pillow.org

Appendix B: Complete Code Listing

The full codebase (∼3,884 lines, 13 files) is at:

Project Links:

- GitHub Repository: https://github.com/Kritnandan22/street-cleanliness-ai
- Live Demo (HuggingFace Spaces): https://huggingface.co/spaces/Nandan22/StreatClean_AI

B.1 Core Pipeline

#	File	Description
1	main.py (460 L)	CLI entry point. Handles image, video, demo commands with argument parsing and result serialization.
2	app.py (164 L)	Flask web application. Drag-and-drop upload, full pipeline execution, quad-split visualization.
3	config.py (92 L)	Central configuration: scene baselines, semantic weights, thresholds, visualization params.

B.2 Inference

#	File	Description
4	inference/detection_pipeline.py (390 L)	Core orchestrator. Loads all models, runs detection + classification, invokes scoring modules, produces final composite report.

B.3 Models

#	File	Description
5	models/scene_classifier.py (271 L)	MobileNetV2 scene classifier with custom 4-class head, ImageNet preprocessing, training utilities.

B.4 Scoring Utilities

#	File	Description
6	utils/context_aware_scorer.py (258 L)	**Novelty 1.** Scene-dependent context normalization. Configurable baselines, score banding, recommendation generation.
7	utils/spatial_heatmap.py (332 L)	**Novelty 2.** Grid-based spatial analysis. 8×8 grid, Gaussian smoothing, hotspot detection, Gini computation, OpenCV visualization.
8	utils/weighted_semantic_scorer.py (315 L)	**Novelty 3.** Ecological-weighted semantic scoring. Configurable class weights, area-normalized contributions, per-class impact breakdown.

B.5 Training and Evaluation

#	File	Description
9	train.py (233 L)	YOLOv8 fine-tuning script. Dataset validation, augmentation config, early stopping, weight export.
10	evaluate.py (483 L)	Ablation study framework. All 4 scoring modes, aggregate statistics, bar charts, CSV export.

B.6 Data Processing

#	File	Description
11	data/dataset_loader.py (327 L)	COCO and YOLO format readers; COCO↔YOLO coordinate transforms; multi-source dataset merger.
12	data/prepare_taco_yolo_subset.py (194 L)	TACO→YOLO preparation. Maps 60 TACO classes to 6 semantic groups, stratified splitting.

B.7 Visualization

#	File	Description
13	visualization/visualizer.py (365 L)	Class-colored bounding boxes, JET heatmap overlay, hotspot circle markers, score HUD, quad-split assembly, ablation chart generation.

Total codebase: ∼3,884 lines across 13 Python source files.

Group Member Contributions

- **Chayan Kumawat - 220309**: Handled dataset preprocessing by consolidating the TACO dataset into 6 semantic classes. Trained the YOLOv8n model for 30 epochs and achieved a performance of mAP@0.5 = 0.144.
- **Mudit Agrawal - 220673**: Developed the Scene Classification module using MobileNetV2 and implemented the Context-Aware Scoring mechanism, which normalizes litter counts based on scene type on a 0–5 scale.
- **Sangam Gupta - 220961**: Designed the Spatial Heatmap module, including an 8×8 Gaussian density grid for hotspot detection and visualization of litter clusters.
- **Karan Jangid - 220500**: Implemented the Ecological Semantic Scorer, incorporating material persistence weighting (e.g., plastic having 128.6× higher persistence than organic waste), and integrated the complete inference pipeline.
- **Kritnandan - 220550**: Developed the Flask-based web dashboard, managed Docker and Hugging Face deployment, and contributed to the CVPR-format report along with experimental visualizations.